1.

```
MariaDB [mynewpaltz]> select firstName, lastName, StudentTable.ssn, CourseTable.title from StudentT
able left join EnrollmentTable on StudentTable.ssn = EnrollmentTable.ssn
    -> left join CourseTable on EnrollmentTable.courseID = CourseTable.courseID
    -> where StudentTable.ssn IN(
    -> select ssn from EnrollmentTable group by ssn having count(courseID) = 5
    -> );
+-----------+----------+-----------+----------------------------+
| firstName | lastName | ssn       | title                      |
+-----------+----------+-----------+----------------------------+
| Kevin     | Green    | 100000041 | Calculus I                 |
| Kevin     | Green    | 100000041 | Calculus II                |
| Kevin     | Green    | 100000041 | Linear Algebra             |
| Kevin     | Green    | 100000041 | Probability and Statistics |
| Kevin     | Green    | 100000041 | Abstract Algebra           |
| Michelle  | Adams    | 100000042 | Intro to Computer Science  |
| Michelle  | Adams    | 100000042 | Data Structures            |
| Michelle  | Adams    | 100000042 | Operating Systems          |
| Michelle  | Adams    | 100000042 | Database Management Systems |
| Michelle  | Adams    | 100000042 | Artificial Intelligence    |
| Brian     | Nelson   | 100000043 | English Composition I      |
| Brian     | Nelson   | 100000043 | English Composition II     |
| Brian     | Nelson   | 100000043 | World Literature           |
| Brian     | Nelson   | 100000043 | Creative Writing           |
| Brian     | Nelson   | 100000043 | Shakespeare                |
| Carol     | Baker    | 100000044 | Intro to Engineering       |
| Carol     | Baker    | 100000044 | Statics                    |
| Carol     | Baker    | 100000044 | Dynamics                   |
| Carol     | Baker    | 100000044 | Thermodynamics             |
| Carol     | Baker    | 100000044 | Circuit Analysis           |
| Edward    | Hall     | 100000045 | Calculus I                 |
| Edward    | Hall     | 100000045 | Linear Algebra             |
| Edward    | Hall     | 100000045 | Abstract Algebra           |
| Edward    | Hall     | 100000045 | Data Structures            |
| Edward    | Hall     | 100000045 | Database Management Systems |
| Ronald    | Campbell | 100000047 | English Composition I      |
| Ronald    | Campbell | 100000047 | World Literature           |
| Ronald    | Campbell | 100000047 | Shakespeare                |
| Ronald    | Campbell | 100000047 | Statics                    |
| Ronald    | Campbell | 100000047 | Thermodynamics             |
| Melissa   | Mitchell | 100000048 | English Composition II     |
| Melissa   | Mitchell | 100000048 | Creative Writing           |
| Melissa   | Mitchell | 100000048 | Intro to Engineering       |
| Melissa   | Mitchell | 100000048 | Dynamics                   |
| Melissa   | Mitchell | 100000048 | Circuit Analysis           |
| Timothy   | Carter   | 100000049 | Calculus I                 |
| Timothy   | Carter   | 100000049 | Probability and Statistics |
| Timothy   | Carter   | 100000049 | Data Structures            |
| Timothy   | Carter   | 100000049 | Artificial Intelligence    |
| Timothy   | Carter   | 100000049 | World Literature           |
| Deborah   | Roberts  | 100000050 | Calculus II                |
| Deborah   | Roberts  | 100000050 | Abstract Algebra           |
| Deborah   | Roberts  | 100000050 | Operating Systems          |
| Deborah   | Roberts  | 100000050 | English Composition I      |
| Deborah   | Roberts  | 100000050 | Creative Writing           |
| Stephanie | Phillips | 100000052 | Probability and Statistics |
| Stephanie | Phillips | 100000052 | Data Structures            |
| Stephanie | Phillips | 100000052 | Artificial Intelligence    |
```

2.

```
MariaDB [mynewpaltz]> select firstName, lastName, StudentTable.ssn, CourseTable.title from StudentT
able left join EnrollmentTable on StudentTable.ssn = EnrollmentTable.ssn
    -> left join CourseTable on EnrollmentTable.courseID = CourseTable.courseID
    -> where StudentTable.ssn IN(
    -> select ssn from EnrollmentTable group by ssn having count(courseID) = 4);
+-----------+----------+-----------+----------------------------+
| firstName | lastName | ssn       | title                      |
+-----------+----------+-----------+----------------------------+
| Amanda    | Rivera   | 100000046 | Calculus II                |
| Amanda    | Rivera   | 100000046 | Probability and Statistics |
| Amanda    | Rivera   | 100000046 | Intro to Computer Science  |
| Amanda    | Rivera   | 100000046 | Operating Systems          |
| Jacob     | Cruz     | 100000057 | Database Management Systems |
| Jacob     | Cruz     | 100000057 | English Composition II     |
| Jacob     | Cruz     | 100000057 | Shakespeare                |
| Jacob     | Cruz     | 100000057 | Dynamics                   |
+-----------+----------+-----------+----------------------------+
8 rows in set (0.001 sec)
```

3.

```
MariaDB [mynewpaltz]> select firstName, lastName, StudentTable.ssn, CourseTable.title from StudentT
able left join EnrollmentTable on StudentTable.ssn = EnrollmentTable.ssn
    -> left join CourseTable on EnrollmentTable.courseID = CourseTable.courseID
    -> where StudentTable.ssn IN(
    -> select ssn from EnrollmentTable group by ssn having count(courseID) = 3);
+-------------+-----------+-----------+----------------------------+
| firstName   | lastName  | ssn       | title                      |
+-------------+-----------+-----------+----------------------------+
| James       | Smith     | 100000001 | Calculus I                 |
| James       | Smith     | 100000001 | Intro to Computer Science  |
| James       | Smith     | 100000001 | English Composition I      |
| Mary        | Johnson   | 100000002 | Calculus II                |
| Mary        | Johnson   | 100000002 | Data Structures            |
| Mary        | Johnson   | 100000002 | English Composition II     |
| Robert      | Williams  | 100000003 | Linear Algebra             |
| Robert      | Williams  | 100000003 | Operating Systems          |
| Robert      | Williams  | 100000003 | World Literature           |
| Patricia    | Brown     | 100000004 | Probability and Statistics |
| Patricia    | Brown     | 100000004 | Database Management Systems |
| Patricia    | Brown     | 100000004 | Creative Writing           |
| John        | Jones     | 100000005 | Abstract Algebra           |
| John        | Jones     | 100000005 | Artificial Intelligence    |
| John        | Jones     | 100000005 | Shakespeare                |
| Jennifer    | Garcia    | 100000006 | Calculus I                 |
| Jennifer    | Garcia    | 100000006 | Intro to Computer Science  |
| Jennifer    | Garcia    | 100000006 | Intro to Engineering       |
| Michael     | Miller    | 100000007 | Calculus II                |
| Michael     | Miller    | 100000007 | Data Structures            |
| Michael     | Miller    | 100000007 | Statics                    |
| Linda       | Davis     | 100000008 | Linear Algebra             |
| Linda       | Davis     | 100000008 | Operating Systems          |
| Linda       | Davis     | 100000008 | Dynamics                   |
| David       | Rodriguez | 100000009 | Probability and Statistics |
| David       | Rodriguez | 100000009 | Database Management Systems |
| David       | Rodriguez | 100000009 | Thermodynamics             |
| Elizabeth   | Martinez  | 100000010 | Abstract Algebra           |
| Elizabeth   | Martinez  | 100000010 | Artificial Intelligence    |
| Elizabeth   | Martinez  | 100000010 | Circuit Analysis           |
| William     | Hernandez | 100000011 | Calculus I                 |
| William     | Hernandez | 100000011 | English Composition I      |
| William     | Hernandez | 100000011 | Intro to Engineering       |
| Barbara     | Lopez     | 100000012 | Calculus II                |
| Barbara     | Lopez     | 100000012 | English Composition II     |
| Barbara     | Lopez     | 100000012 | Statics                    |
| Richard     | Gonzalez  | 100000013 | Linear Algebra             |
| Richard     | Gonzalez  | 100000013 | World Literature           |
| Richard     | Gonzalez  | 100000013 | Dynamics                   |
| Susan       | Wilson    | 100000014 | Probability and Statistics |
| Susan       | Wilson    | 100000014 | Creative Writing           |
| Susan       | Wilson    | 100000014 | Thermodynamics             |
| Joseph      | Anderson  | 100000015 | Abstract Algebra           |
| Joseph      | Anderson  | 100000015 | Shakespeare                |
| Joseph      | Anderson  | 100000015 | Circuit Analysis           |
| Jessica     | Thomas    | 100000016 | Calculus I                 |
| Jessica     | Thomas    | 100000016 | Intro to Computer Science  |
| Jessica     | Thomas    | 100000016 | English Composition I      |
| Thomas      | Taylor    | 100000017 | Calculus II                |
```

4.

```
    -> select ssn from EnrollmentTable group by ssn having count(courseID) = 2);
+-----------+------------+-----------+----------------------------+
| firstName | lastName   | ssn       | title                      |
+-----------+------------+-----------+----------------------------+
| Nicholas  | Stewart    | 100000061 | Abstract Algebra           |
| Nicholas  | Stewart    | 100000061 | Artificial Intelligence    |
| Kathleen  | Morris     | 100000062 | Intro to Computer Science  |
| Kathleen  | Morris     | 100000062 | English Composition I      |
| Eric      | Morales    | 100000063 | Data Structures            |
| Eric      | Morales    | 100000063 | English Composition II     |
| Amy       | Murphy     | 100000064 | Operating Systems          |
| Amy       | Murphy     | 100000064 | World Literature           |
| Stephen   | Cook       | 100000065 | Database Management Systems |
| Stephen   | Cook       | 100000065 | Creative Writing           |
| Shirley   | Rogers     | 100000066 | Artificial Intelligence    |
| Shirley   | Rogers     | 100000066 | Shakespeare                |
| Jonathan  | Gutierrez  | 100000067 | English Composition I      |
| Jonathan  | Gutierrez  | 100000067 | Intro to Engineering       |
| Angela    | Ortiz      | 100000068 | English Composition II     |
| Angela    | Ortiz      | 100000068 | Statics                    |
| Larry     | Morgan     | 100000069 | World Literature           |
| Larry     | Morgan     | 100000069 | Dynamics                   |
| Helen     | Cooper     | 100000070 | Creative Writing           |
| Helen     | Cooper     | 100000070 | Thermodynamics             |
| Justin    | Peterson   | 100000071 | Shakespeare                |
| Justin    | Peterson   | 100000071 | Circuit Analysis           |
| Anna      | Bailey     | 100000072 | Calculus I                 |
| Anna      | Bailey     | 100000072 | Intro to Engineering       |
| Scott     | Reed       | 100000073 | Calculus II                |
| Scott     | Reed       | 100000073 | Statics                    |
| Brenda    | Kelly      | 100000074 | Linear Algebra             |
| Brenda    | Kelly      | 100000074 | Dynamics                   |
| Brandon   | Howard     | 100000075 | Probability and Statistics |
| Brandon   | Howard     | 100000075 | Thermodynamics             |
| Pamela    | Ramos      | 100000076 | Abstract Algebra           |
| Pamela    | Ramos      | 100000076 | Circuit Analysis           |
| Frank     | Kim        | 100000077 | Calculus I                 |
| Frank     | Kim        | 100000077 | Intro to Computer Science  |
| Nicole    | Foster     | 100000078 | Calculus II                |
| Nicole    | Foster     | 100000078 | Data Structures            |
| Benjamin  | Alvarado   | 100000079 | Linear Algebra             |
| Benjamin  | Alvarado   | 100000079 | Operating Systems          |
| Emma      | Washington | 100000080 | Probability and Statistics |
| Emma      | Washington | 100000080 | Database Management Systems |
| Gregory   | Butler     | 100000081 | Abstract Algebra           |
| Gregory   | Butler     | 100000081 | Artificial Intelligence    |
| Samantha  | Simmons    | 100000082 | Intro to Computer Science  |
| Samantha  | Simmons    | 100000082 | English Composition I      |
| Samuel    | Foster     | 100000083 | Data Structures            |
| Samuel    | Foster     | 100000083 | English Composition II     |
| Katherine | Gonzales   | 100000084 | Operating Systems          |
| Katherine | Gonzales   | 100000084 | World Literature           |
| Raymond   | Bryant     | 100000085 | Database Management Systems |
| Raymond   | Bryant     | 100000085 | Creative Writing           |
| Christine | Alexander  | 100000086 | Artificial Intelligence    |
| Christine | Alexander  | 100000086 | Shakespeare                |
| Patrick   | Russell    | 100000087 | English Composition I      |
```

5.

```
    -> select ssn from EnrollmentTable group by ssn having count(courseID) = 1);
+-----------+----------+-----------+----------------------------+
| firstName | lastName | ssn       | title                      |
+-----------+----------+-----------+----------------------------+
| Charles   | Lee      | 100000021 | Calculus I                 |
| Nancy     | Perez    | 100000022 | Calculus II                |
| Matthew   | Thompson | 100000023 | Linear Algebra             |
| Lisa      | White    | 100000024 | Probability and Statistics |
| Anthony   | Harris   | 100000025 | Abstract Algebra           |
| Betty     | Sanchez  | 100000026 | Intro to Computer Science  |
| Mark      | Clark    | 100000027 | Data Structures            |
| Margaret  | Ramirez  | 100000028 | Operating Systems          |
| Donald    | Lewis    | 100000029 | Database Management Systems |
| Sandra    | Robinson | 100000030 | Artificial Intelligence    |
| Steven    | Walker   | 100000031 | English Composition I      |
| Ashley    | Young    | 100000032 | English Composition II     |
| Paul      | Allen    | 100000033 | World Literature           |
| Dorothy   | King     | 100000034 | Creative Writing           |
| Andrew    | Wright   | 100000035 | Shakespeare                |
| Kimberly  | Scott    | 100000036 | Intro to Engineering       |
| Joshua    | Torres   | 100000037 | Statics                    |
| Emily     | Nguyen   | 100000038 | Dynamics                   |
| Kenneth   | Hill     | 100000039 | Thermodynamics             |
| Donna     | Flores   | 100000040 | Circuit Analysis           |
+-----------+----------+-----------+----------------------------+
20 rows in set (0.003 sec)
```

6.

```
MariaDB [mynewpaltz]> select firstName, lastName, StudentTable.ssn, CourseTable.title from StudentT
able left join EnrollmentTable on StudentTable.ssn = EnrollmentTable.ssn
    -> left join CourseTable on EnrollmentTable.courseID = CourseTable.courseID
    -> where StudentTable.ssn IN(
    -> select ssn from EnrollmentTable group by ssn having count(courseID) = 0);
Empty set (0.002 sec)
```

7, 8, 9.

```
MariaDB [mynewpaltz]> select count(*) from CourseTable where numOfCredits = 4;
+----------+
| count(*) |
+----------+
|        6 |
+----------+
1 row in set (0.001 sec)

MariaDB [mynewpaltz]> select count(*) from CourseTable where numOfCredits = 3;
+----------+
| count(*) |
+----------+
|       13 |
+----------+
1 row in set (0.001 sec)

MariaDB [mynewpaltz]> select count(*) from CourseTable where numOfCredits = 2;
+----------+
| count(*) |
+----------+
|        1 |
+----------+
1 row in set (0.001 sec)
```

10. No

```
MariaDB [mynewpaltz]> select * from CourseTable where numOfCredits = 5;
Empty set (0.001 sec)

MariaDB [mynewpaltz]> select * from CourseTable where numOfCredits = 1;
Empty set (0.001 sec)
```

11.

```
MariaDB [mynewpaltz]> select CourseTable.title, firstName, lastName, Enrollment
Table.grade from StudentTable left join EnrollmentTable on StudentTable.ssn = E
nrollmentTable.ssn
    -> left join CourseTable on EnrollmentTable.courseID = CourseTable.courseID

    -> order by CourseTable.title;
+----------------------------+-------------+------------+-------+
| title                      | firstName   | lastName   | grade |
+----------------------------+-------------+------------+-------+
| NULL                       | Jack        | Myers      | NULL  |
| NULL                       | Jason       | Gomez      | NULL  |
| Abstract Algebra           | Elizabeth   | Martinez   | A     |
| Abstract Algebra           | Joseph      | Anderson   | A     |
| Abstract Algebra           | Karen       | Martin     | C     |
| Abstract Algebra           | Edward      | Hall       | A     |
| Abstract Algebra           | Maria       | Wallace    | B     |
| Abstract Algebra           | Anthony     | Harris     | B     |
| Abstract Algebra           | Jeffrey     | Evans      | B     |
| Abstract Algebra           | Gregory     | Butler     | B     |
| Abstract Algebra           | Kevin       | Green      | B     |
| Abstract Algebra           | Nicholas    | Stewart    | A     |
| Abstract Algebra           | Deborah     | Roberts    | C     |
| Abstract Algebra           | John        | Jones      | C     |
| Abstract Algebra           | Pamela      | Ramos      | C     |
| Artificial Intelligence    | Michelle    | Adams      | A     |
| Artificial Intelligence    | Gregory     | Butler     | B     |
| Artificial Intelligence    | Stephanie   | Phillips   | C     |
| Artificial Intelligence    | Sharon      | Edwards    | C     |
| Artificial Intelligence    | Christine   | Alexander  | C     |
| Artificial Intelligence    | Nicholas    | Stewart    | B     |
| Artificial Intelligence    | John        | Jones      | A     |
| Artificial Intelligence    | Elizabeth   | Martinez   | C     |
| Artificial Intelligence    | Timothy     | Carter     | C     |
| Artificial Intelligence    | Karen       | Martin     | B     |
| Artificial Intelligence    | Shirley     | Rogers     | B     |
| Artificial Intelligence    | Sandra      | Robinson   | A     |
| Artificial Intelligence    | Ryan        | Diaz       | C     |
| Calculus I                 | Timothy     | Carter     | A     |
| Calculus I                 | Tyler       | Woods      | A     |
| Calculus I                 | Edward      | Hall       | B     |
| Calculus I                 | Anna        | Bailey     | B     |
| Calculus I                 | James       | Smith      | A     |
| Calculus I                 | Jennifer    | Garcia     | B     |
| Calculus I                 | William     | Hernandez  | B     |
| Calculus I                 | Jessica     | Thomas     | C     |
| Calculus I                 | Charles     | Lee        | A     |
| Calculus I                 | Janet       | Ford       | A     |
| Calculus I                 | Cynthia     | Reyes      | A     |
| Calculus I                 | Kevin       | Green      | A     |
| Calculus I                 | Frank       | Kim        | C     |
| Calculus II                | Heather     | Cole       | C     |
| Calculus II                | Scott       | Reed       | C     |
| Calculus II                | Kevin       | Green      | B     |
| Calculus II                | Dennis      | Hamilton   | B     |
| Calculus II                | Deborah     | Roberts    | B     |
| Calculus II                | Amanda      | Rivera     | A     |
| Calculus II                | Mary        | Johnson    | B     |
```

12.

```
MariaDB [mynewpaltz]> select title, courseNumber, subjectID, count(StudentTable
.firstName) as studentCount
    -> from CourseTable left join EnrollmentTable on CourseTable.courseID = Enr
ollmentTable.courseID
    -> left join StudentTable on EnrollmentTable.ssn = StudentTable.ssn
    -> where CourseTable.courseID = 10;
+-------------------------+--------------+-----------+--------------+
| title                   | courseNumber | subjectID | studentCount |
+-------------------------+--------------+-----------+--------------+
| Artificial Intelligence |          450 | CS        |           13 |
+-------------------------+--------------+-----------+--------------+
1 row in set (0.006 sec)
```

13.

```
MariaDB [mynewpaltz]> select courseID, count(ssn) as numOfStudents from Enrollm
entTable group by courseID;
+----------+---------------+
| courseID | numOfStudents |
+----------+---------------+
|        1 |            13 |
|        2 |            13 |
|        3 |            12 |
|        4 |            14 |
|        5 |            13 |
|        6 |            11 |
|        7 |            13 |
|        8 |            13 |
|        9 |            12 |
|       10 |            13 |
|       11 |            14 |
|       12 |            13 |
|       13 |            14 |
|       14 |            14 |
|       15 |            12 |
|       16 |            12 |
|       17 |            12 |
|       18 |            12 |
|       19 |            11 |
|       20 |            10 |
+----------+---------------+
20 rows in set (0.002 sec)
```

14.

```
MariaDB [mynewpaltz]> select firstName, lastName, sum(CourseTable.numOfCredits)
 as credits from StudentTable
    -> join EnrollmentTable on StudentTable.ssn = EnrollmentTable.ssn
    -> join CourseTable on EnrollmentTable.courseID = CourseTable.courseID
    -> group by StudentTable.ssn, StudentTable.firstName, StudentTable.lastName
;
+-------------+------------+---------+
| firstName   | lastName   | credits |
+-------------+------------+---------+
| James       | Smith      |      11 |
| Mary        | Johnson    |      10 |
| Robert      | Williams   |      10 |
| Patricia    | Brown      |       9 |
| John        | Jones      |       9 |
| Jennifer    | Garcia     |      10 |
| Michael     | Miller     |      10 |
| Linda       | Davis      |      10 |
| David       | Rodriguez  |      10 |
| Elizabeth   | Martinez   |      10 |
| William     | Hernandez  |       9 |
| Barbara     | Lopez      |      10 |
| Richard     | Gonzalez   |       9 |
| Susan       | Wilson     |      10 |
| Joseph      | Anderson   |      10 |
| Jessica     | Thomas     |      11 |
| Thomas      | Taylor     |      10 |
| Sarah       | Moore      |      10 |
| Christopher | Jackson    |       9 |
| Karen       | Martin     |       9 |
| Charles     | Lee        |       4 |
| Nancy       | Perez      |       4 |
| Matthew     | Thompson   |       3 |
| Lisa        | White      |       3 |
| Anthony     | Harris     |       3 |
| Betty       | Sanchez    |       4 |
| Mark        | Clark      |       3 |
| Margaret    | Ramirez    |       4 |
| Donald      | Lewis      |       3 |
| Sandra      | Robinson   |       3 |
| Steven      | Walker     |       3 |
| Ashley      | Young      |       3 |
| Paul        | Allen      |       3 |
| Dorothy     | King       |       3 |
| Andrew      | Wright     |       3 |
| Kimberly    | Scott      |       2 |
| Joshua      | Torres     |       3 |
| Emily       | Nguyen     |       3 |
| Kenneth     | Hill       |       4 |
| Donna       | Flores     |       4 |
| Kevin       | Green      |      17 |
| Michelle    | Adams      |      17 |
| Brian       | Nelson     |      15 |
| Carol       | Baker      |      16 |
| Edward      | Hall       |      16 |
| Amanda      | Rivera     |      15 |
| Ronald      | Campbell   |      16 |
| Melissa     | Mitchell   |      15 |
```